

$$\underline{1.1(4)} \quad \frac{(1+2i)^2 - (1-i)^3}{(3+2i)^3 - (2+i)^2} = \frac{1+4i-4-1+3i+3-i}{27+54i-36-8i-4-4i+1} =$$

$$= \frac{-1+6i}{-2+42i} = \frac{(-1+6i)(42i+12)}{-42^2-12^2} = \frac{44-5i}{318}$$

$$\underline{1.5(4)} \quad \bar{z}^8 = 1+i$$

$$|z|e^{i\varphi} = \sqrt{2}e^{i\frac{\pi}{4}}$$

$$\begin{cases} |z|^8 = \sqrt{2} \\ 8\varphi = \frac{\pi}{4} + 2\pi k \end{cases} \Rightarrow \begin{cases} |z| = \sqrt[8]{2} \\ \varphi = \frac{\pi}{32} + \frac{\pi k}{4} \end{cases} \Rightarrow \bar{z} = \sqrt[8]{2} e^{i(\frac{\pi}{32} + \frac{\pi k}{4})}$$

1.6. $P(\bar{z}_0) = 0$; $\forall \lambda \in \mathbb{R}, \bar{z} \in \mathbb{C}$

Правда из чл-х комплексных чисел $\begin{cases} \overline{\lambda z} = \lambda \bar{z} \\ \overline{\bar{z}^n} = z^n \end{cases}$

Тогда $\overline{P(\bar{z})} = P(\bar{z}_0) = 0$

1.10(5)

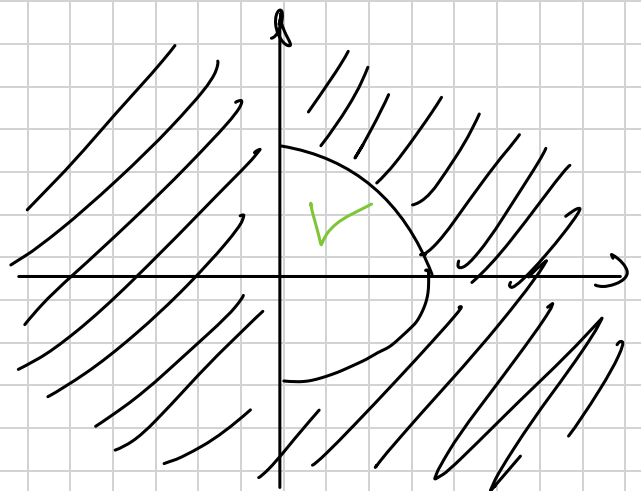
$$0 < \arg \frac{i-z}{1+z} < \frac{\pi}{2}$$

$$\frac{i-z}{z+1} = w, \quad \begin{cases} \operatorname{Re} w > 0 \\ \operatorname{Im} w > 0 \end{cases}$$

$$w = \frac{i-x-iy}{x+iy+1} =$$

$$= \frac{(i(1-y)-x)(x-i(y+1))}{(x+i(1-y))(x-i(y+1))} = \frac{-x^2-y^2+1+i[x(y+1)+x(1-y)]}{x^2+(y+1)^2} \quad \Rightarrow$$

$$\Rightarrow \begin{cases} (1-y^2) - x^2 > 0 \\ (1-y)x + x(y+1) > 0 \end{cases} \Rightarrow \begin{cases} x^2 + y^2 < 1 \\ 2x > 0 \end{cases}$$



1.11. $A|z|^2 + \bar{B}z + B\bar{z} + C = 0$ $A, C \in \mathbb{R}, B \in \mathbb{C}, A > 0, AC < |B|^2$

$$|z|^2 + \frac{\bar{B}}{A}z + \frac{B}{A}\bar{z} + \frac{C}{A} = 0 \Rightarrow$$

$$\Rightarrow x^2 + y^2 + \frac{(\alpha - i\beta)(x + iy)}{A} + \frac{(\alpha + i\beta)(x - iy)}{A} + \frac{C}{A} = 0 \Rightarrow$$

$$\Rightarrow x^2 + y^2 + \frac{2}{A}(\alpha x + \beta y) + \frac{C}{A} = 0 \Rightarrow$$

$$\Rightarrow \left(x + \frac{\alpha}{A}\right)^2 + \left(y + \frac{\beta}{A}\right)^2 = \frac{\alpha^2}{A^2} + \frac{\beta^2}{A^2} - \frac{C}{A} = \frac{|B|^2 - AC}{A^2}$$

Тогда центр $-\frac{B}{A}$, радиус $-\sqrt{\frac{|B|^2 - AC}{A^2}}$

1.16. $\exists a_i \in \mathbb{R}, a_i - \forall P(z) = \bar{z}^n + a_1 \bar{z}^{n-1} + \dots + a_n$ уел.

и действ. корней. ~~Сказано~~, что $Q(z) = P(z+ia) + P(z-ia)$ уел. и действ. корни.

$$P(z) = (z - c_1)(z - c_2) \dots (z - c_n) \quad c_i - \text{корни.}$$

$$Q(z) = (z - c_1 + ia)(z - c_2 + ia) \dots + (z - c_1 - ia) \dots = 0$$

$$|(z - c_1 + ia) \dots (z - c_n + ia)|^2 = |(z - c_1 - ia) \dots (z - c_n - ia)(-1)|^2$$

$$|(z - c_k)|^2 = |x + yi - c_k + ia|^2 = (x - c_k)^2 + (a + y)^2$$

$$|(z - c_k)|^2 = (x - c_k)^2 + (y - a)^2 \Rightarrow y = 0 \Rightarrow \text{Все корни } Q(z) \text{ действ.}$$

2.1.4) $z_n = \frac{1}{\sqrt{n}} (1 - e^{i\varphi} + \dots + (-1)^n e^{in\varphi}) = \frac{1}{\sqrt{n}} \sum_{k=0}^n e^{i\pi} e^{ik\varphi} =$

$$= \frac{1}{\sqrt{n}} \sum_{k=0}^n e^{i(\pi + \varphi)k} = \frac{1}{\sqrt{n}} \sum_{k=0}^n e^{i\alpha k} = \frac{1}{\sqrt{n}} \frac{e^{i\alpha(n+1)} - 1}{e^{i\alpha} - 1}$$

$$\left| \frac{e^{i\alpha(n+1)} - 1}{e^{i\alpha} - 1} \right| \leq \frac{2}{e^{i\alpha} - 1} \text{ из леммы } \Delta \Rightarrow \frac{1}{\sqrt{n}} \rightarrow 0, n \rightarrow \infty$$

$$\Rightarrow z_n \rightarrow 0, n \rightarrow \infty$$

2.12. ζ -л, эо сфрмкостн на сфре Римана отвает на комплексной плоскости или окр. или прямая окр. на сфре Римана

$$\begin{cases} Ax + By + Cz + D = 0 \\ x^2 + y^2 + (z - \frac{1}{2})^2 = \frac{1}{4} \end{cases}$$

NA || NB :

$$NA = \begin{pmatrix} x-0 \\ y-0 \\ z-0 \end{pmatrix} \quad NB = \begin{pmatrix} x-0 \\ y-0 \\ 0-1 \end{pmatrix}$$

$$\frac{x}{y} = \frac{z}{-z} = \frac{z-1}{-z} \Rightarrow x = \frac{z}{1-z} \quad y = \frac{z}{1-z}$$

$$x = x(x, y)$$

$$x^2 + y^2 + z(z-1) = 0 \Rightarrow x = x(1-z) \Rightarrow$$

$$y = y(x, y)$$

$$y = y(1-z)$$

$$z = z(x, y)$$

$$\Rightarrow x^2(1-z)^2 + y^2(1-z)^2 - z(1-z)$$

$$\frac{z}{1-z} = x^2 + y^2 \Rightarrow z = \frac{x^2 + y^2}{x^2 + y^2 + 1} ; \quad x = \frac{x}{x^2 + y^2 + 1} ; \quad y = \frac{y}{x^2 + y^2 + 1}$$

$$Ax + By + C(x^2 + y^2) + D(x^2 + y^2 + 1) \Rightarrow (x^2 + y^2)(C + D) + Ax + By + D = 0$$

3.12(2) $\sin(iy) = \frac{e^{-y} + e^y}{2i} = i \operatorname{sh} y ; \quad \cos(iy) = \operatorname{ch} y$

$$\cos z = \cos x \cos(iy) - \sin x \sin(iy) = \cos x \operatorname{ch} y - i(\sin x \operatorname{sh} y) \Rightarrow$$

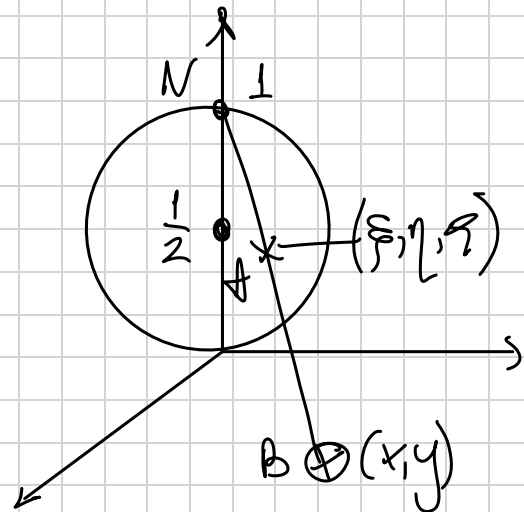
$$\Rightarrow \operatorname{Re} \cos z = \cos x \operatorname{ch} y$$

$$\operatorname{Im} \cos z = -\sin x \operatorname{sh} y$$

3.13(2) $\sin z = \sin x \operatorname{ch} y + i(\cos x \operatorname{sh} y) \Rightarrow$

$$\Rightarrow |\sin z| = \sqrt{\sin^2 x \operatorname{ch}^2 y + \cos^2 x \operatorname{sh}^2 y} = \sqrt{\operatorname{ch}^2 y - \sin^2 x - 1} =$$

$$= \sqrt{\operatorname{ch}^2 y - \cos^2 x}$$



$$\underline{3.17(3)} \quad \cos z = \frac{3i}{4} \Rightarrow \frac{e^{iz} - e^{-iz}}{2} = \frac{3i}{4} \Rightarrow [w = e^{iz}] \Rightarrow$$

$$\Rightarrow 2w + 2w^{-1} = 3i \Rightarrow 2w^2 - 3wi + 2 = 0 \Rightarrow$$

$$\Rightarrow \begin{cases} w = 2i \\ w = -\frac{1}{2}i \end{cases} \Rightarrow \begin{cases} iz = \ln(2i) = \ln(2) + i(\frac{\pi}{2} + 2\pi k) \\ iz = \ln(-\frac{1}{2}i) = \ln(\frac{1}{2}) + i(-\frac{\pi}{2} + 2\pi k) \end{cases}$$

$$4) \quad \cos z = \frac{3+i}{4} \Rightarrow 2w + 2w^{-1} = 3+i \Rightarrow \begin{cases} w = 1+i \\ w = \frac{1}{2}(1-i) \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} z = \frac{\pi}{4} + 2\pi k - i \cdot \frac{1}{2} \ln 2 \\ z = -\frac{\pi}{4} + 2\pi k + i \cdot \frac{1}{2} \ln 2 \end{cases}$$

$$\underline{3.19.} \quad 1) \quad e^z = e^x (\cos y + i \sin y)$$

$|e^z| = e^x \rightarrow \infty, x \rightarrow \infty$ - растет по y в any $\mathbb{R}$.

аналогично $e^x \rightarrow 0, x \rightarrow -\infty$ - p. по y

$$3) \quad |\sin z| = \sqrt{\cosh^2 y - \cos^2 x} \geq \sqrt{\cosh^2 y - 1} \rightarrow \infty, y \rightarrow \pm\infty$$

растет по x , аналогично $\cos z$

$$\underline{4.6(2)} \quad \sum_{n=1}^{\infty} n^3 e^{-nz} \quad \operatorname{Re} z \geq \delta > 0$$

$$|n^3 e^{-n(\operatorname{Re} z + i \operatorname{Im} z)}| = |n^3 e^{-n \operatorname{Re} z}| |e^{-n i \operatorname{Im} z}| \leq |n^3 e^{-n \delta}| \rightarrow 0$$

$$\frac{(n+1)^3 e^{-(n+1)\delta}}{n^3 e^{-n\delta}} = \left(1 + \frac{1}{n}\right)^3 \cdot C < 1 \Rightarrow \text{Вейерштрасс}$$

$$\underline{4.7(2)} \quad \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = 1 \quad \sum_{n=1}^{\infty} a_n e^{-nz} \quad \text{Запрос } \varepsilon < \delta$$

$$|a_n e^{-nz}| = |a_n| e^{-nx} \leq e^{-n(\delta - \varepsilon)} \quad \text{выберем } \varepsilon \text{ так, } n$$

$$|a_n| \leq (1 + \varepsilon)^n = e^{n\varepsilon(1+\varepsilon)} \leq e^{\varepsilon n}$$

При этом ряд $\sum_{n=1}^{\infty} e^{-n(\delta - \varepsilon)}$ - с. $\Rightarrow$ по пр. Вейерштр. с. $\checkmark$ исход.

5.1.4) $z = x + iy$ $f(z) = (\bar{z})^2$ $f(z) = (x^2 + y^2 - 2ixy)$

$f = u(x, y) + i v(x, y)$ $\text{условие Коши-Гoursa: } \begin{cases} u_x = v_y \\ v_x = -u_y \end{cases}$

$$\begin{cases} u_x = 2x \\ v_y = -2x \end{cases} \quad \begin{cases} v_x = -2y \\ u_y = 2y \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases}$$

5.7.(6) $f(z) = \frac{\sin z}{\sin z - \cos z}$. При $\sin z \neq \cos z$

~~мы~~ $\text{найти производную: } f'(z) = \frac{-1}{(\sin z - \cos z)^2}$

$\frac{e^{iz} - e^{-iz}}{2i} = \frac{e^{iz} + e^{-iz}}{2}$. Пусть $t = e^{iz} \Rightarrow t^2 / (1 - i) = 1 + i$

$t^2 = \frac{\sqrt{2} e^{i\pi/4}}{\sqrt{2} e^{-i\pi/4}} = e^{i\pi/2} \Rightarrow t = e^{i\pi/4 + i\pi k} = e^{iz} \Rightarrow$

$\Rightarrow \ln(i\pi/4 + i\pi k) = \ln(i\pi/4) \Rightarrow z = \frac{\pi}{4} + \pi k, k \in \mathbb{Z}$

5.11. $u^2 + v^2 = 0$ $\Delta u = 0$ $\Delta v = 0$ реш. совпадающие

ген. к-д: $\begin{cases} u_x = v_y \\ u_y = -v_x \end{cases}$ $\Phi\text{-ф: } \ln(u^2 + v^2) - \text{реш.}$

$\frac{\partial u}{\partial x} = \frac{1}{u^2 + v^2} \cdot \left(2u \frac{\partial u}{\partial x} + 2v \frac{\partial v}{\partial x} \right)$

$\frac{\partial^2 u}{\partial x^2} = -\frac{1}{(u^2 + v^2)^2} \left(2u \frac{\partial u}{\partial x} + 2v \frac{\partial v}{\partial x} \right)^2 +$

$+ \frac{1}{u^2 + v^2} \left(2 \left(\frac{\partial u}{\partial x} \right)^2 + 2u \frac{\partial^2 u}{\partial x^2} + 2 \left(\frac{\partial v}{\partial x} \right)^2 + 2v \frac{\partial^2 v}{\partial x^2} \right)$

$\Delta u = \frac{-4}{(u^2 + v^2)^2} \left(u^2 \left(\frac{\partial u}{\partial x} \right)^2 + v^2 \left(\frac{\partial v}{\partial x} \right)^2 + 2uv \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + u^2 \left(\frac{\partial u}{\partial y} \right)^2 + \right.$
 $\left. + v^2 \left(\frac{\partial v}{\partial y} \right)^2 + 2uv \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} \right) + \frac{2}{u^2 + v^2} \left(\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right)$

5.17.7) $|f(z)| = (x^2 + y^2)e^x = |ze^z| \Rightarrow \underline{f(z) = ze^z}$

$$\Rightarrow f(z) = (x+iy)e^x(\cos y + i\sin y) = xe^x\cos y - ye^x\sin y + i(ye^x\cos y + xe^x\sin y)$$

Узн. к-р. :
$$\begin{cases} u_x = e^x\cos y + xe^x\cos y - ye^x\cos y = V_y \\ u_y = -xe^x\sin y - e^x\sin y - ye^x\cos y = -V_x \end{cases} \quad \checkmark$$

Узн. нрм. : $0 = \Delta u = 2e^x\cos y + xe^x\cos y - ye^x\sin y - xe^x\cos y - 2e^x\cos y + ye^x\sin y$
 $\Delta v = -2e^x\sin y - ye^x\cos y - xe^x\cos y + ye^x\cos y + 2e^x\sin y + xe^x\sin y = 0$

8) $\arg f(z) = 2xy$

$$f(z) = |f(z)|e^{2ixy} \Rightarrow \underline{f(z) = e^{x^2+y^2}e^{2ixy} = e^{z^2}}$$

Узн. к-р. :
$$\begin{cases} u_x = 2xe^{x^2+y^2}\cos(2xy) - e^{x^2+y^2}\sin(2xy)2y = V_y \\ u_y = -2ye^{x^2+y^2}\cos(2xy) - e^{x^2+y^2}\sin(2xy)2x = -V_x \end{cases}$$

Узн. нрм. - проверен в Wolfram - правильно

7.1 $f(z) = 2|xy| + i|x^2 - y^2|$

Проверен узн. кр в 4 случаях:

1) $xy > 0, x^2 - y^2 > 0 \Rightarrow u_x \neq V_y$

2) $xy > 0, x^2 - y^2 < 0 \Rightarrow u_x = V_y, u_y = -V_x \quad \checkmark$

3) $xy < 0, x^2 - y^2 < 0 \Rightarrow u_x \neq V_y$

4) $xy < 0, x^2 - y^2 > 0 \Rightarrow u_x = V_y, u_y = -V_x \quad \checkmark$

7.2 $f(z) = u(x,y) + iv(x,y)$ - per. $\Rightarrow \begin{cases} u_x = v_y \\ u_y = -v_x \end{cases}$

$|grad u| = |grad v|$ uz th. Rucanove odobijeno

7.2(4) $\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n \quad |z| < 1$

$$\frac{1}{(1+z^2)^2} = \sum_{n=0}^{\infty} (n+1)(-1)^n z^{2n}$$

$$\frac{1}{1+u} = \sum_{k=0}^{\infty} (-1)^k u^k \quad |u| < 1$$

$$\frac{-1}{(1+u)^2} = \sum_{k=0}^{\infty} (-1)^k k u^{k-1}$$

$$\frac{1}{(1+u)^2} = \sum_{k=1}^{\infty} (-1)^{k+1} k u^{k-1} = \sum_{n=0}^{\infty} (-1)^n (n+1) u^n = \sum_{n=0}^{\infty} (-1)^n (n+1) z^{2n}$$

7.3.18) $\frac{z^3(z+1)}{(z^2+1)(z-1)(z+1)} = \frac{z^3(z+1)}{z^4-1} = -(z^4+z^3) \cdot \frac{1}{1-z^4} =$
 $= (-z^4+z^3) \sum_{k=0}^{\infty} z^{4k} = \sum_{k=0}^{\infty} -z^{4k+4} + \sum_{k=0}^{\infty} -z^{4k+3}$

13) $f(z) = \frac{1}{(1-z^4)(1+z+z^2+z^3)} = \frac{1-z}{(1-z^4)^2} = (1-z) \sum_{n=0}^{\infty} (n+1) z^{4n} =$
 $= \sum_{n=0}^{\infty} (n+1) (z^{4n} - z^{4n+1})$

7.4. $F(z) = \frac{2z^2+2z-z}{z^2+z-2} \quad z_0 = -1$

$$F(z) = 2 - \frac{3}{z^2+z-2} = 2 - \frac{1}{z-1} + \frac{1}{z+2} = \left[\begin{array}{l} u = z+1 \\ z = u-1 \end{array} \right]$$

$$= 2 - \frac{1}{u-2} + \frac{1}{u+1} = 2 - \frac{1}{2} \frac{1}{u/2-1} + \frac{1}{u+1} =$$

$$= 2 + \sum_{k=0}^{\infty} \left(\frac{1}{2} \left(\frac{u}{2} \right)^k + (-1)^k u^k \right) = \frac{7}{2} + \sum_{k=1}^{\infty} \left(\frac{1}{2^{k+1}} + (-1)^k \right) (z+1)^k$$

$$\underline{7.6.(3)} \quad f(z) = \sin^4 z + \cos^4 z = (\sin^2 z + \cos^2 z)^2 - 2\sin^2 z \cos^2 z = \\ = 1 - \frac{1}{2} \sin^2(2z) = \frac{3}{4} + \frac{\cos 4z}{4} = \frac{3}{4} + \sum_{n=0}^{\infty} (-1)^n \frac{2^{4n-2} z^{2n}}{(2n)!}$$

$$\underline{7.6.6)} \quad f(z) = \operatorname{ch} z \cos z = \frac{e^z + e^{-z}}{2} \cdot \frac{e^{iz} + e^{-iz}}{2} = \\ = \frac{e^{z(1+i)} + e^{-z(1+i)}}{4} + \frac{e^{z(1-i)} + e^{-z(1-i)}}{4} = \frac{1}{2} \operatorname{ch}(1+i)z + \frac{1}{2} \operatorname{ch}(1-i)z = \\ = \frac{1}{2} \sum_{k=0}^{\infty} \frac{((1+i)z)^{2k}}{(2k)!} + \frac{1}{2} \sum_{k=0}^{\infty} \frac{((1-i)z)^{2k}}{(2k)!} = \\ = \sum_{k=0}^{\infty} \frac{1}{2} \frac{2^k (e^{i\frac{\pi}{2}k} - e^{-i\frac{\pi}{2}k})}{(2k)!} z^{2k} = \sum_{k=0}^{\infty} \frac{2^k}{(2k)!} \cos \frac{\pi}{2} k z^{2k} = \\ = \sum_{k=0}^{\infty} \frac{2^{2k}}{(4k)!} (-1)^k z^{4k}$$

$$\underline{7.7} \quad f(z) = (z^2 - 4z + 5)e^{4z - z^2}, \quad z_0 = 2$$

$$f(t) = ((t+2)^2 - 4t - 3)e^{4t+8 - (t+2)^2} = (t^2 + 1)e^{-t^2+4} = \\ = \sum_{n=0}^{\infty} \frac{(-1)^n e^4}{n!} (t^{2n+2} + t^{2n}) \Rightarrow f(z) = \sum_{n=0}^{\infty} \frac{(-1)^n e^4}{n!} ((z-2)^{2n+2} + (z-2)^{2n})$$

$$\underline{7.12(1)} \quad f'(z) = f(z), \quad f(0) = 1 \Rightarrow f^{(k)}(0) = f(0) = 1 \Rightarrow \\ \Rightarrow f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

$$\underline{T3} \quad 1) \quad 3i = e^{i(\ln 3 + 2\pi k)} = e^{i \ln 3 - 2\pi k} = e^{-2\pi k} (\cos(\ln 3) + i \sin(\ln 3))$$

$$2) \quad i = e^{i(\ln i + i(\frac{\pi}{2} + 2\pi k))} = e^{-\frac{\pi}{2} - 2\pi k}$$

$$3) \quad (-1)^i = e^{\ln(-1)^i} = e^{i \ln(-1)} = e^{-\pi + 2\pi k}$$

$$\ln z = \ln |z| + i(\arg z + 2\pi k);$$

$$\underline{9.2(5)} \quad f\left(\frac{1}{n}\right) = \frac{1}{2n + \cos \pi n} \left(= \frac{1}{2n + (-1)^n} \right)$$

$$\exists f(z) \text{ per.}, \quad z_n = \frac{1}{n} \rightarrow 0, n \rightarrow \infty$$

$$z_{2k} = \frac{1}{2k} \quad f\left(\frac{1}{2k}\right) = \frac{1}{(4k+1)} = \frac{1/2k}{2 + \frac{1}{2k}} \Rightarrow f(z_{2k}) = \frac{z_{2k}}{2 + z_{2k}}$$

$$g(z) = \frac{z}{2+z}; \quad \left(f(z) - g(z) \right) \Big|_{z=\frac{1}{2k}} = 0$$

$$\downarrow \text{per.} \\ \text{even per., } \forall \quad f(z) \equiv g(z)$$

$$g\left(\frac{1}{2k+1}\right) = \frac{1}{2(2k+1)+1} \quad \neq \quad \Rightarrow \text{not}$$

$$f\left(\frac{1}{2k+1}\right) = \frac{1}{2(2k+1)-1}$$

7)

$$\exists \text{ не } f(z) \text{ per. в окрестности } z=0 : f\left(\frac{1}{n}\right) = f\left(-\frac{1}{n}\right) = \frac{1}{2n+1}$$

$$\text{Нужно } \exists f(z) \text{ per., пока } n \rightarrow \infty \quad z_n = \frac{1}{n}$$

$$f(z_n) = f\left(\frac{1}{n}\right) = \frac{z_n}{2+z_n}$$

$$\text{Посм. } g(z) = \frac{z}{2+z}, \quad f - g(z_n) = 0 \Rightarrow g \text{ - per.}, \text{ even } f \text{ - per., } \forall \quad \overbrace{f \equiv g}$$

$$f\left(-\frac{1}{n}\right) = g\left(-\frac{1}{n}\right) = \frac{-\frac{1}{n}}{2-\frac{1}{n}} = \frac{1}{1-2n} \neq \frac{1}{2n+1} \Rightarrow \text{not}$$

8)

$$\exists \text{ не } f(z) \text{ - per.} \quad f\left(\frac{1}{n}\right) = e^{-n}$$

$$\text{Нужно } \exists f(z) \text{ - per., пока } f(z) = \sum_{k=0}^{\infty} a_k z^k$$

$$f\left(\frac{1}{n}\right) = \sum_{k=0}^{\infty} a_k \frac{1}{n^k} = e^{-n}; \quad \lim_{n \rightarrow \infty} \left(a_0 + \sum_{k=1}^{\infty} a_k \left(\frac{1}{n}\right)^k \right) = a_0 + 0 = \underbrace{\lim_{n \rightarrow \infty} e^{-n}}_{=0} = 0$$

$$f'(z) = \sum a_k k z^{k-1}$$

$$f'\left(\frac{1}{n}\right) = \sum_{k=1}^{\infty} a_k k \left(\frac{1}{n}\right)^{k-1} = a_1 + \sum_{k=2}^{\infty} a_k n \left(\frac{1}{n}\right)^{k-1}$$

$$\left\{ \begin{array}{l} f'\left(\frac{1}{n}\right) = \sum a_k k \left(\frac{1}{n}\right)^{k-1} \rightarrow a_1, n \rightarrow \infty \\ n f'\left(\frac{1}{n}\right) = n e^{-n} \Rightarrow a_1 = 0 \end{array} \right.$$

$$\left\{ \begin{array}{l} f''\left(\frac{1}{n}\right) \rightarrow a_2, n \rightarrow \infty \\ n^2 f\left(\frac{1}{n}\right) = e^{-n} n^2 \Rightarrow a_2 = 0 \end{array} \right. \Rightarrow a_i = 0 \Rightarrow f\left(\frac{1}{n}\right) = 0 \neq e^{-n}$$

з.з. Ныяр $f(z)$ - пер. б $z=0$ а не const

$$z_k = \frac{1}{2^k} \Rightarrow f(z_k) = \text{const. Тогор сему нэ факт}$$

$$n\text{-ту перем } (f(z) - \underbrace{c(z)}_{\text{const}}) \equiv 0, c(z) - \text{per}$$

Сему $f(z)$ - пер., то $f(z) \equiv \text{const} \Rightarrow$ такот ф. нэ

14 $f(z) = f(z_0) + \sum_{k=1}^{n-1} c_k (z-z_0)^k$ - пер. б Теленора

Сему перем. такот $g(z)$, то $f(z) - g(z) \equiv 0$ б уаре, не $f(z)$ - пер., ну эом $g(z)$ - нонетом с. $< n \Rightarrow f$ нэ

11.1.1) $\sum_{n=0}^{\infty} 2^{-n} z^n = \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n + \underbrace{\sum_{n=1}^{\infty} 2^n z^n}_{\sum_{n=1}^{\infty} 2^n z^{-n}}$

$$\frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{1}{2}\right)^n}} = 2 - \text{паруе с.}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{2^n} = \frac{1}{2} \Rightarrow \text{консго с.} \quad \underbrace{1/2 < |z| < 2}$$

$$11.3.3) \quad f(z) = \frac{z}{(z^2+1)(z+2)} = \frac{2z+1}{5} \frac{1}{1+z^2} - \frac{1}{5} \frac{1}{1+\frac{z}{2}} \quad \ominus$$

$$[Az^2 + (2A+B)z + 2B + C \quad z^2 + C = z]$$

$$\ominus - \frac{2z+1}{5} \sum_{n=0}^{\infty} (-z)^{2n} - \frac{1}{5} \sum_{n=0}^{\infty} \left(\frac{-z}{2}\right)^n = \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{5} (2z^{2n+1} + z^{2n})$$

В конуре $1 < |z| < 2$ во exp. степенях

$$11.4, \quad f(z) = \frac{z^2-1}{z^2+1} \quad \text{Разложим на } z^{-1} \quad z_0 = 2i$$

$$\frac{z^2-1}{z^2+1} = 1 - \frac{2}{z^2+1} = 1 + \frac{A}{z+i} + \frac{B}{z-i}; \Rightarrow f(z) = 1 - \frac{i}{z+i} + \frac{i}{z-i}$$

$$f(u) = 1 - \frac{i}{u+i+1} + \frac{i}{u+1-i}$$

$$z_0 = 2i; \quad u_0 = 2i+1 \quad |u_0| = \sqrt{5}$$

$$\frac{i}{u+1-i} = \frac{i}{u} \left(\frac{1}{1 + \frac{1-i}{u}} \right) =$$

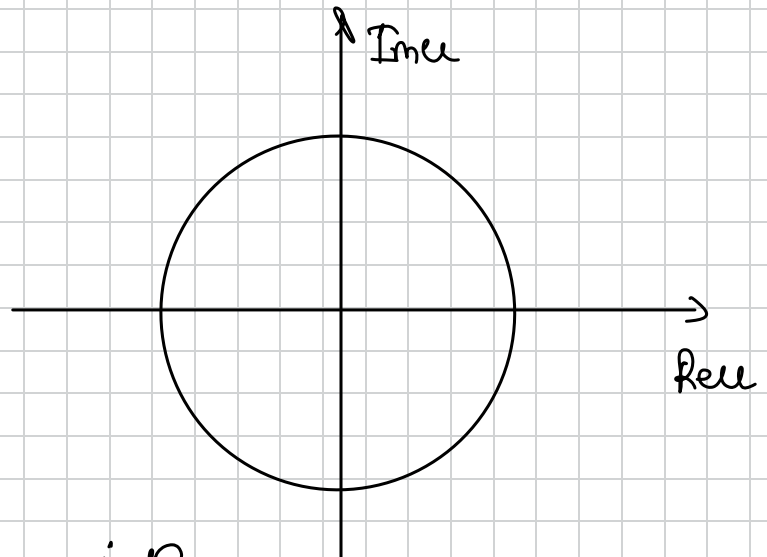
$$= \frac{i}{u} \left(\sum_{n=0}^{\infty} (-1)^n \left(\frac{1-i}{u} \right)^n \right) =$$

$$= \sum_{n=0}^{\infty} \frac{i(-1)^n (e^{i\frac{3\pi}{4}})^n (2)^n}{u^{n+1}} =$$

$$= \sum_{n=1}^{\infty} \frac{i(-1)^{n-1}}{u^n} (1-i)^{n-1} = \sum_{n=1}^{\infty} \frac{i(-1)^{n-1}}{(z-1)^n} (1-i)^{n-1}$$

$$\sum_{n=1}^{\infty} \frac{-i(-1)^{n-1}}{(z-1)^n} (1+i)^{n-1} = \frac{-i}{u+i+1}$$

$$\text{Other: } f(z) = 1 + \sum_{n=1}^{\infty} \frac{i(-1)^{n-1} [(1-i)^{n-1} - (1+i)^{n-1}]}{(z-1)^n} \quad \text{при } \sqrt{2} < |z-1|$$



Ряды сходимости:

$$|u| < |1-i| = \sqrt{2}$$

$$\sqrt{2} < u < +\infty$$

$$11.5(5) \quad f(z) = \frac{3z+1}{z(z+1)} - \frac{z}{z^2+5} = \left(3 + \frac{1}{z}\right) \frac{1}{1+z} - \frac{z}{5} \cdot \frac{1}{1+\frac{z^2}{5}}$$

Konvergenz: $0 < |z| < \frac{1}{2}$, $\frac{1}{2} < |z| < \sqrt{5}$, $|z| > \sqrt{5}$

$$z_0 = 1 \Rightarrow |z_0| = 1$$

$$\begin{aligned} f(z) &= \left(3 + \frac{1}{z}\right) \sum_{n=0}^{\infty} (-z)^n - \frac{z}{5} \sum_{n=0}^{\infty} \left(\frac{z^2}{5}\right)^n = \\ &= \sum_{n=0}^{\infty} \left(-\frac{1}{5}\right)^{n+1} z^{2n+1} + \sum_{n=0}^{\infty} (-1)^{n+1} \cdot (2z)^n + \frac{3}{2z} \quad \text{b. } \frac{1}{2} < |z| < \sqrt{5} \quad \text{Konvergenz} \end{aligned}$$

$$11.9(2) \quad z_0 = -1-5i, \quad t = z+1$$

$$f(z) = \frac{-4-2i}{(z+1+2i)(z-3)} = \frac{1}{z+1+2i} - \frac{1}{z-3} \Rightarrow f(t) = \frac{1}{2i+t} - \frac{1}{t-4}$$

Konvergenz: $|t| < 2$, $2 < |t| < 4$, $|t| > 4$, $t_0 = -5i \Rightarrow |t_0| = 5$

$$f(t) = -\frac{1}{2i} \sum_{n=0}^{\infty} \left(\frac{-t}{2i}\right)^n - \frac{1}{4} \sum_{n=0}^{\infty} \left(\frac{t}{4}\right)^n \quad \text{b. sicher } |t| > 4$$

$$11.10(7) \quad z_0 = -1-i, \quad t = z+3+5i$$

$$f(z) = \frac{16-z^2}{z(z+4i)^2} = -\frac{1}{z} + \frac{8i}{(z+4i)^2} \Rightarrow f(t) = -\frac{1}{t-3-5i} + \frac{8i}{(t-3-5i)^2}$$

Konvergenz: $|t| < \sqrt{10}$, $\sqrt{10} < |t| < \sqrt{34}$, $|t| > \sqrt{34}$

$$t_0 = 2+4i \Rightarrow |t_0| = 2\sqrt{5}$$

$$f(t) = \frac{1}{3+5i} \sum_{n=0}^{\infty} \left(\frac{t}{3+5i}\right)^n + \frac{8i}{(3+5i)^2} \sum_{n=0}^{\infty} n \left(\frac{t}{3+5i}\right)^{n+1} \quad \text{b. } \sqrt{10} < |t| < \sqrt{34} \quad \text{Konvergenz}$$

$$8(4) \quad \frac{1}{z^2-1} \cdot \cos\left(\frac{\pi z}{z+1}\right) = f(z)$$

$$z = \infty, \quad z+1=0, \quad z^2+1=0$$

$$1) \quad z = \infty, \quad f(z) \sim -\frac{1}{z^2} \xrightarrow{z \rightarrow \infty} 0 \Rightarrow \text{устранимая}$$

$$2) \quad z = -1 \quad \text{где} \quad \frac{1}{z^2-1} = \infty^1 \quad \text{где} \quad \cos \frac{\pi z}{z+1} - \text{не имеет смысла}$$

$$3) \quad z = 1 - \text{устр}, \text{ т.к. } f(z) \sim \frac{\sin\left(\frac{\pi}{2} \left(\frac{z-1}{z+1}\right)\right)}{z^2-1} \sim \frac{-\pi}{2(z+1)^2} = -\frac{\pi}{8} \quad z \rightarrow 1$$

8(7) 1) исходные точки $z = \infty$, нули всех знаменателей, устр

2) исходные точки

критич. осн. т.

$$e^{i\pi/z}$$

$$1) \quad z = \infty, \quad z = 0, \quad \sin \frac{\pi}{z} = 0 \quad z_n = \frac{1}{k}, \quad k \in \mathbb{Z}$$

$$2) \quad z = 0 - \text{устр}$$

$z = \infty$: $\lim_{z \rightarrow \infty} f(z) \Rightarrow$ устр. особая точка

$$g(z) = e^{f(z)} \quad z_0 - \text{нуль } f(z)$$

$$f(z) \approx \frac{h(z)}{(z-z_0)^k} \quad f(z) = \frac{h(z)}{|z-z_0|^k} \quad \arg f(z) = \arg h - k \arg(z-z_0)$$

$$f = |f| e^{i \arg f} \quad z_n \rightarrow z_0 \quad \arg h - k \arg(z_n - z_0) = 0, \quad e^{f(z)} \rightarrow +\infty$$

$$z_m \rightarrow z_0 \quad \arg h - k \arg(z_m - z_0) = \pi, \quad e^{f(z)} \rightarrow 0$$

$$z_k = \frac{1}{k} - \text{устр. осн.}$$

$$12.15(1) \quad f(z) = \frac{1 - \operatorname{ch} \frac{z}{2}}{e^z - e^{3z}}$$

$$\text{Критич. осн.}: \quad z = \infty \Rightarrow e^z = e^{3z}$$

$$\Rightarrow \begin{cases} x = 0 \\ y = 3y + 2\sqrt{x} \end{cases} \Rightarrow z = \pi k i$$

$$z_k \rightarrow \infty, k \rightarrow \infty \Rightarrow z = \infty - \text{точ}$$

Рассмотрим, где нулями оф. в 0: $\operatorname{ch} \frac{z}{2} = 1 \Rightarrow z_k = 4\pi k$

$$1 - \operatorname{ch} \frac{z}{2} = 1 - \operatorname{ch} \frac{z - z_{k1}}{2} \sim \frac{(z - z_{k1})^2}{4}, \quad z \rightarrow z_{k1}$$

В этом $\sim (z - z_{k1}) \Rightarrow$ год, т.к. $\exists$ конечный $\lim f(z)$

В остальных точках год. Π^1

$$\underline{12.15(11)} \quad f(z) = \frac{4z^2 + 48z - 8\pi^2}{(e^{2iz} - 1)^2} - \frac{3\pi}{z + 2\pi}$$

Качество: $z = \infty, z = -2\pi, e^{2iz} = 1 \Rightarrow z_k = \pi k$

$z = \infty$ - точ, т.к. $z_k \rightarrow \infty, k \rightarrow \infty$

Рассмотрим нули оф. в 0 в $-2\pi, \pi$, и в расм. отсюда

$$z = \pi: f(t) = \frac{2t \cdot (t + 3\pi)}{-4t^2} - \frac{3\pi}{t + 3\pi} \sim t^{-1} \Rightarrow \Pi^1$$

$$z = -2\pi \Rightarrow f(t) = \frac{4(t - 3\pi)t}{-4t^2} - \frac{3\pi}{t} = -1 + \frac{3\pi}{t} - \frac{3\pi}{t} \sim -1 \Rightarrow \text{год}$$

В ост. точках: Π^2

$$\underline{12.17.5)} \quad f(z) = \frac{8z^2 - 6z + 1}{1 - \cos \frac{\pi}{z}} \quad z = \infty \quad z = 0 \quad \frac{\pi}{z} = 2\pi k \quad \frac{1}{z_k} = \frac{1}{2k}$$

1) z - точ, т.к. $z_k \rightarrow 0, k \rightarrow \infty$

2) z_k - кратные нули знаменателя

$$(1 - \cos \frac{\pi}{z})' = \sin \frac{\pi}{z} \left(\frac{-\pi}{z^2} \right) - \text{точ оф. в 0}$$

Нули числителя: $z = \frac{1}{4}, z = \frac{1}{2}$

17 Zuerst, so $|z^3 f(z)| \leq B|z| \Rightarrow$

$\Rightarrow$ наблр. з. дога наблр. соединит те done тем 2 снрг.

Proof $\forall \alpha > 0$ $\exists \epsilon = 0 - n^1$, $\alpha = 0 \rightarrow \text{yes}$ $\alpha = 1 \rightarrow n^2$
 $\alpha \neq 1 \rightarrow n^1$